

Weight Subspace Superposition: A Unified Geometry of Low-Rank Training Dynamics

DRAFT

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1 Contributions

We present a unifying weight parameterization scheme for low memory parameter efficient pretraining of LLMs through superpositioned subspace factors of W on $\text{St}(r,n)$. We show that these have attractive geometry, and allow for reduced memory and parameter efficient pretraining of LLMs as well as clean analysis of several open questions in the interpretability of LLMs.

1. **Theoretical unification:** The weight subspace superposition theory unifies prior work in activation and gradient subspace optimization methods through a clean factorization of the subspaces of weights. [show that WASI and GrassWalk are limits of WSS theory, connect to MoE somehow?]
2. **Parameter Efficient LLM Pretraining:** Lmao maybe
3. **How many Heads?:** Measuring the von Neumann entropy of the subspaces of W provides a clean estimate for how many factorization states $j \in J$ are needed. In a simple case, this is analogous to the number of heads in multihead attention.
4. **Applications to Mechanistic Interpretability:** The factorization of sample X contributions to ΔW and A across J tractable $r \times n$ subspaces provides a natural delineation between potentially semantically interpretable feature axes. Further, these reduced rank spaces allow for direct computation of H^{-1} , critical for accurate influence function estimation ($I(X, \theta) \approx \delta\theta$). [so much more to think about here - lots of difficult math]
5. **Data Efficiency:** The proposed gating function $\phi(u)$ is a natural inroad to identifying high and low utility data points in the training set. This is deeply connected to the influence function contribution alluded to above. [connections to IG and MoE work]

2 Setup and notation

For a layer with weight $W \in \mathbb{R}^{n \times m}$, input $X \in \mathbb{R}^{B \times n}$, and bias $b \in \mathbb{R}^m$, the forward map is $A = XW + \mathbf{1}b^\top$. We approximate W at each layer $i \in \{1, \dots, I\}$ by a sum of J rank- r components,

$$W_i \approx \widetilde{W}_i = \frac{1}{J} \sum_{j=1}^J f_{i,j}(X) U_{i,j} \Sigma_{i,j} V_{i,j}^\top, \quad (1)$$

where, suppressing the layer index i where unambiguous:

$$\begin{aligned} U_j &\in \text{St}(r, n) \quad (\text{left frame, orthonormal columns}), \\ V_j &\in \text{St}(r, m) \quad (\text{right frame, orthonormal columns}), \\ \Sigma_j &\in \mathbb{R}_{\text{diag}, > 0}^{r \times r} \quad (\text{spectrum}), \\ f_{i,j}(X) &\in \mathbb{R} \quad (\text{content gate, see §4}). \end{aligned}$$

The forward pass becomes

$$\tilde{A}_i = \frac{1}{J} \sum_{j=1}^J f_{i,j}(X) (X U_{i,j}) \Sigma_{i,j} V_{i,j}^\top + b_i, \quad (2)$$

which is computed in $O(JB(n+m)r)$ time per layer, never materializing the $n \times m$ matrix $\widetilde{W}_i$.

3 Geometry

We use three manifolds with non-overlapping roles.

Stiefel $\text{St}(r, n)$. $\text{St}(r, n) = \{U \in \mathbb{R}^{n \times r} : U^\top U = I_r\}$ is the manifold of ordered orthonormal r -frames in $\mathbb{R}^n$, with $\dim = nr - \binom{r+1}{2}$. The optimizer state lives here: U_j and V_j are updated as matrices.

Grassmann $\text{Gr}(r, n)$. $\text{Gr}(r, n) = \text{St}(r, n)/\text{O}(r)$, identifying $U \sim UR$ for $R \in \text{O}(r)$. Subspaces, not frames, with $\dim = r(n-r)$. The diversity penalty (§9) is a function on $\text{Gr}(r, n)^J$: it depends only on $\text{col}(U_j)$, not on the choice of basis. Equivalently, it depends only on the orthogonal projector $P_j = U_j U_j^\top$.

Fixed-rank $\mathcal{M}_r(n, m)$. $\mathcal{M}_r(n, m) = \{X \in \mathbb{R}^{n \times m} : \text{rank} X = r\}$, with $\dim = (n+m)r - r^2$. The data-fit loss sees $X_j = U_j \Sigma_j V_j^\top \in \mathcal{M}_r(n, m)$ and is invariant to the $\text{O}(r)$ gauge $U_j \mapsto U_j R$, $V_j \mapsto V_j R$ (jointly).

Working setup. Parameters are stored on the product manifold

$$\mathcal{P} = \prod_{j=1}^J \left[\text{St}(r, n) \times \text{St}(r, m) \times \mathbb{R}_{> 0}^r \right],$$

with one such factor per layer. The optimizer takes Riemannian steps on the Stiefel factors and Euclidean steps on the spectrum factors (with a positivity barrier or reparametrization $\Sigma = \exp(\text{diag}(s))$ to keep $\Sigma \succ 0$).

4 The content gate

A linear gate $f_{i,j} \in \mathbb{R}$ that does not depend on X leaves (1) equivalent to a single rank- Jr factorization: any decomposition of a rank- Jr target into J rank- r pieces achieves the same loss. The $J > 1$ structure is then unidentified and offers no expressive gain over $J = 1$ with rank Jr .

To make $J > 1$ a strict generalization, we choose a *content-addressed* gate that activates each component according to how much of the input lies in that component’s right subspace:

$$f_{i,j}(X) = \phi\left(\frac{\|V_{i,j}^\top X^\top\|_F^2}{\|X\|_F^2}\right), \quad (3)$$

where $\phi : [0, 1] \rightarrow \mathbb{R}$ is a fixed (or learnable-scalar-parametrized) nonlinear scalar function. Reasonable choices include $\phi(u) = u$ (linear in subspace energy), $\phi(u) = \sigma(\alpha u + \beta)$ (sigmoid with learned scale/shift), or $\phi(u) = u^\gamma$ for $\gamma > 0$.

This gate has three properties we want:

- (i) *Subspace-aware.* It ties component activation to the geometry $V_{i,j}$, not to an independent learned router. Two components with identical right subspaces receive identical gates, eliminating the collapse mode where $V_{i,j} = V_{i,k}$ becomes a free direction in parameter space.
- (ii) *Input-dependent.* The forward map is now nonlinear in $\widetilde{W}_i$ when viewed as a function of X : different inputs activate different mixtures of components, so the layer is no longer a fixed linear map. This is the nonlinearity that justifies $J > 1$.
- (iii) *Cheap.* The numerator $\|V_{i,j}^\top X^\top\|_F^2 = \text{tr}(V_{i,j}^\top X^\top X V_{i,j})$ requires only the $B \times r$ matrix $X V_{i,j}$, which is already computed in the forward pass (2). Cost: $O(Br)$ extra per component.

Remark 1. (3) is one concrete choice; the more general principle is $f_{i,j}$ depends on X through the geometry of component j . Asymmetric variants (gate on U instead of V , or on both) are reasonable alternatives. A more exotic option would be represent $f_{i,j}(X)$ with a *LayerNorm* or *Linear layer* which adapts over training, which would require correct gradient flow through ϕ .

5 Forward and backward

Write $H_j := X U_j \in \mathbb{R}^{B \times r}$ and $K_j := X^\top X V_j \in \mathbb{R}^{n \times r}$. Let $g_j := f_{i,j}(X)$ be the realized gate values for a minibatch.

Forward.

$$\widetilde{A}_i = \frac{1}{J} \sum_j g_j H_j \Sigma_j V_j^\top + b_i. \quad (4)$$

Backward. Let $\delta := \partial L / \partial \widetilde{A}_i \in \mathbb{R}^{B \times m}$. The data-fit gradient through component j , treating g_j as fixed (we address the gate gradient below), is

$$\frac{\partial L}{\partial U_j} = \frac{1}{J} g_j X^\top (\delta V_j \Sigma_j^\top), \quad (5)$$

$$\frac{\partial L}{\partial V_j} = \frac{1}{J} g_j \delta^\top (H_j \Sigma_j), \quad (6)$$

$$\frac{\partial L}{\partial \Sigma_j} = \frac{1}{J} g_j \text{diag}(H_j^\top \delta V_j). \quad (7)$$

Components decouple: the gradient w.r.t. (U_j, Σ_j, V_j) depends on other components only through the shared upstream signal δ . This is the parallelization we want; the chain rule does the work. These expressions are expanded in 9 for the subspace diversity measured through von Neumann entropy on U_j and V_j . They should not change much, but are important notes for implementation.

Gate gradient. From (3), g_j depends on V_j (and only V_j). Writing $u_j := \|V_j^\top X^\top\|_F^2 / \|X\|_F^2$,

$$\frac{\partial g_j}{\partial V_j} = \frac{2\phi'(u_j)}{\|X\|_F^2} X^\top X V_j = \frac{2\phi'(u_j)}{\|X\|_F^2} K_j. \quad (8)$$

The full $\partial L / \partial V_j$ is the sum of (6) and the gate-induced term

$$\left. \frac{\partial L}{\partial V_j} \right|_{\text{gate}} = \left(\frac{\partial L}{\partial g_j} \right) \frac{\partial g_j}{\partial V_j}, \quad \frac{\partial L}{\partial g_j} = \frac{1}{J} \langle \delta, H_j \Sigma_j V_j^\top \rangle_F.$$

The inner product is computed without materializing the outer product as $\text{tr}(\Sigma_j H_j^\top \delta V_j)$, an $r \times r$ trace.

Cost summary. Per step per layer, dominated by J component contractions of size $B \times (n+m) \times r$. Total $O(JB(n+m)r)$ for forward, same order for backward. No $n \times m$ object is ever formed.

6 Riemannian step

For $U_j \in \text{St}(r, n)$ with Euclidean gradient $G_j := \partial L / \partial U_j$ (combining all sources, data-fit and gate and diversity), the canonical-metric Riemannian gradient is

$$\hat{G}_j = G_j - U_j \text{sym}(U_j^\top G_j), \quad \text{sym}(M) := \frac{1}{2}(M + M^\top). \quad (9)$$

We use the QR retraction

$$U_j^+ = \text{qf}(U_j - \eta \hat{G}_j), \quad (10)$$

where qf extracts the orthonormal factor from a thin QR decomposition. This is a second-order approximation to the geodesic and costs $O(nr^2)$ per component. The same procedure applies to $V_j \in \text{St}(r, m)$.

The Σ_j factors update by standard Euclidean SGD/Adam on the diagonal, with the positivity reparametrization $\Sigma_j = \exp(\text{diag}(s_j))$ if desired.

Remark 2 (Retraction choice). *The QR retraction is one of three standard options on $\text{St}(r, n)$. The polar retraction $R_{\text{pol}}(U, \eta) = (U - \eta \hat{G})((U - \eta \hat{G})^\top (U - \eta \hat{G}))^{-1/2}$ has the same $O(nr^2)$ cost (with larger constant) and is second-order accurate to the geodesic; it has no sign ambiguity but is otherwise comparable. The geodesic (exponential map) is exact along the manifold but requires a matrix exponential of a $(2r) \times (2r)$ block, with substantially larger constants. For small step sizes and first-order optimizers, all three retractions give equivalent convergence rates (Absil–Mahony–Sepulchre; Boumal); the choice is constant-factor only. Consequences of future adoption of geodesic stepping are mentioned briefly throughout, but the assumption is that a first build will probably use QR retraction in the interest of time.*

7 Optimizer state under Riemannian updates

Adam-style optimizers maintain first- and second-moment state M_t, V_t per parameter. For Euclidean parameters this is type-safe: state, parameter, and gradient share the same vector space. For $U_j \in \text{St}(r, n)$ moving along the manifold, the tangent space at the new iterate U_j^+ differs from the tangent space at U_j , so momentum from the previous step lives in the wrong vector space. Three approaches:

(A) Projection-based vector transport. After each retraction, project the moment estimates onto the new tangent space. For the canonical metric on $\text{St}(r, n)$,

$$\mathcal{P}_{T_U \text{St}}(\Xi) = \Xi - U \text{sym}(U^\top \Xi). \quad (11)$$

Applied to M_j after retraction, this gives Riemannian Adam in the sense of Bécigneul–Ganea (2019). Cost: $O(nr^2)$ per parameter per step, same order as the retraction itself.

(B) Ambient state with projected gradient. Store M_j, V_j as unconstrained $n \times r$ matrices, update them with standard Adam rules using the Riemannian gradient $\widehat{G}_j$, and project only when computing the update direction. Theoretically less clean than (A) but practically robust and what GrassWalk-style methods do for the analogous gradient-subspace problem.

(C) Reset on retraction. Zero the moment estimates each step. In practice this is not a serious option, especially for LLM pretraining settings.

I recommend (A) for the Stiefel factors U_j, V_j and standard Euclidean Adam for the spectra Σ_j . Concretely, with Riemannian gradient $\widehat{G}_j^{(t)} = G_j^{(t)} - U_j^{(t)} \text{sym}((U_j^{(t)})^\top G_j^{(t)})$:

$$M_j^{(t)} \leftarrow \beta_1 M_j^{(t-1)} + (1 - \beta_1) \widehat{G}_j^{(t)}, \quad (12)$$

$$V_j^{(t)} \leftarrow \beta_2 V_j^{(t-1)} + (1 - \beta_2) (\widehat{G}_j^{(t)})^{\odot 2}, \quad (13)$$

$$U_j^{(t+1)} \leftarrow \text{qf}\left(U_j^{(t)} - \eta \widehat{M}_j^{(t)} / (\sqrt{\widehat{V}_j^{(t)}} + \epsilon)\right), \quad (14)$$

$$M_j^{(t)} \leftarrow \mathcal{P}_{T_{U_j^{(t+1)}} \text{St}}(M_j^{(t)}), \quad (15)$$

where $\widehat{M}, \widehat{V}$ are the bias-corrected moments.

Caveat on the second moment. V_j is element-wise nonlinear in G and does not transport cleanly between tangent spaces. The projection above is an approximation. A more principled treatment in the gradient-subspace setting is given by LDAdam-style corrections (Robert et al., 2024) [check if Claude hallucinated this]; the analog for our weight-side parametrization is open. In practice, projecting both M and V at each step and absorbing the resulting error as optimizer noise works.

Residual gauge in the optimizer. The Stiefel projection $\widehat{G}_j = G_j - U_j \text{sym}(U_j^\top G_j)$ removes the right- $O(r)$ gauge acting on U_j alone, but the parametrization has a residual coupled gauge $(U_j, V_j, \Sigma_j) \mapsto (U_j R, V_j R, R^\top \Sigma_j R)$ that the per-factor projection does not address. Momentum accumulating in this direction has no effect on $\widehat{W}_i$ but inflates effective step sizes. The per-component re-canonicalization in the following section eliminates this for free, though.

8 Periodic re-factorization

Two failure modes motivate periodic intervention on the factor representation. The first is benign and continuous; the second is rare and discrete.

Per-component re-canonicalization (every step? Every few steps?). Within each component, the parametrization $U_j \Sigma_j V_j^\top$ has a residual $O(r)$ gauge that the Riemannian step does not eliminate. Over many steps, Σ_j drifts away from diagonal form in the chosen basis. To restore canonical SVD form, compute the small $r \times r$ SVD $\Sigma_j = R_U S R_V^\top$ and absorb:

$$U_j \leftarrow U_j R_U, \quad V_j \leftarrow V_j R_V, \quad \Sigma_j \leftarrow S. \quad (16)$$

This is exact (the component matrix $U_j \Sigma_j V_j^\top$ is unchanged) and costs $O(r^3 + nr^2 + mr^2)$ per component. The optimizer state must be rotated correspondingly: $M_j^U \leftarrow M_j^U R_U$ and analogously for V .

We should do this at high frequency to reduce momentum accumulation in the coupled gauge spaces of $\text{St}(r, n)$ and $\text{Gr}(r, n)$, which has the effect of increasing step sizes. Swapping to geodesics may partially alleviate the need for this correction step, but they come with more expensive calculations.

Global re-factorization (expensive, rare). The diversity penalty discourages but does not prevent two components from spanning similar subspaces, particularly under unlucky initialization or after long training in a stationary regime. Periodically, materialize and re-factor:

- (1) Compute a randomized SVD of $\widetilde{W}_i = (1/J) \sum_j f_{i,j} U_j \Sigma_j V_j^\top$ to rank Jr , exploiting the factored form. The stacked matrix $[U_1 \Sigma_1, \dots, U_J \Sigma_J] \in \mathbb{R}^{n \times Jr}$ together with the corresponding right factors admits an SVD at cost $O((n+m)(Jr)^2 + (Jr)^3)$ without ever forming the $n \times m$ matrix.
- (2) Redistribute the top Jr singular triplets into J rank- r components. Two reasonable choices: (i) random orthogonal mixing within the top- Jr subspace followed by block partition into J groups of r , which preserves the span exactly and keeps components symmetric; (ii) ordered partition (top r singular vectors as component 1, next r as component 2, etc.), which privileges component ordering and is simpler.
- (3) Reset optimizer state for the re-factored components ($M_j, V_j \leftarrow 0$). The few-step regression from cold momentum is typically invisible at the schedule frequency we recommend.

Schedule. We don't have any good heuristics or data for this choice yet, but this is the basic sketch as I see it:

- Per-component re-canonicalization: every small number of steps.
- Global re-factorization: less frequently, possibly with the period increasing as training progresses. Trigger an off-schedule re-factorization if a diagnostic fires — e.g., $\text{ENC}_i < J/2$, or the smallest pairwise principal angle between components falls below a threshold.

The cost of global re-factorization is non-trivial but amortizes to negligible at the low frequency. The per-component re-canonicalization is essentially free and provides near continuous insurance against gauge drift.

When to skip global re-factorization. If empirical diagnostics show ENC_i remaining near J throughout training and pairwise principal angles staying bounded away from zero, the global step is unnecessary. The per-component step should always be retained.

9 Diversity via density matrices

We can promote subspace diversity through a coordinate-free penalty. At each layer i , define the left and right subspace density matrices

$$\rho_i^L = \frac{1}{J} \sum_{j=1}^J U_{i,j} U_{i,j}^\top, \quad \rho_i^R = \frac{1}{J} \sum_{j=1}^J V_{i,j} V_{i,j}^\top. \quad (17)$$

Both are positive semidefinite with trace r and rank at most Jr , viewed as mixed states over rank- r projectors. We rescale to unit trace, $\bar{\rho} := \rho/r$, and define the diversity penalty as the negative von Neumann entropy

$$\mathcal{D}_i = -S(\bar{\rho}_i^L) - S(\bar{\rho}_i^R), \quad S(\bar{\rho}) := -\text{tr}(\bar{\rho} \log \bar{\rho}). \quad (18)$$

S is maximized when the J subspaces are mutually orthogonal (with $Jr \leq n$), giving $S(\bar{\rho}) = \log(Jr)$, and minimized when all J subspaces coincide, giving $S(\bar{\rho}) = \log r$.

Cheap computation. Stack $\mathbf{U} := [U_1, \dots, U_J] \in \mathbb{R}^{n \times Jr}$. Then

$$\rho^L = \frac{1}{J} \mathbf{U} \mathbf{U}^\top,$$

and the nonzero eigenvalues of ρ^L coincide with those of the smaller $Jr \times Jr$ Gram matrix

$$\mathbf{G} := \frac{1}{J} \mathbf{U}^\top \mathbf{U}.$$

Eigendecomposing $\mathbf{G}$ costs $O((Jr)^3 + nJ^2r^2)$, which is dominated by the per-step optimizer cost for reasonable J .

Gradient of S . For unit-trace $\bar{\rho}$,

$$\frac{\partial S(\bar{\rho})}{\partial \bar{\rho}} = -(I + \log \bar{\rho}), \quad (19)$$

which we push back through $\bar{\rho} = (Jr)^{-1} \sum_j U_j U_j^\top$ by chain rule, yielding

$$\frac{\partial S(\bar{\rho}^L)}{\partial U_j} = -\frac{2}{Jr} (I + \log \bar{\rho}^L) U_j. \quad (20)$$

The matrix logarithm is applied in the eigenbasis of $\bar{\rho}^L$ at cost $O(nJ^2r^2)$ using the eigendecomposition above. The same applies to V_j and $\bar{\rho}^R$.

Effective number of active components. The exponential entropy $\text{ENC}_i := \exp(S(\bar{\rho}_i))/r$ is a continuous proxy for “how many components are effectively active,” with $\text{ENC}_i \in [1, J]$. This is a useful diagnostic and a potential signal for adaptive J .

Training So, incorporation of $\mathcal{D}_i = -\sum_{L,R} S(\bar{\rho}_i)$ through,

$$\frac{\partial S(\bar{\rho}_i^L)}{\partial U_j} = -\frac{2}{Jr} (I + \log(\bar{\rho}_i^L)) U_j \quad (21)$$

And,

$$\frac{\partial S(\bar{\rho}_i^R)}{\partial U_j} = -\frac{2}{Jr} (I + \log(\bar{\rho}_i^R)) V_j \quad (22)$$

We can incorporate these smoothly in the backward pass by saying that at each layer $i \in L$ the loss $\mathcal{L}_i = \bar{\mathcal{L}}_i + \lambda \mathcal{D}_i$, where $\bar{\mathcal{L}}_i$ is the core prediction loss at layer i and λ is some diversity weighting coefficient.

From this view, the gradients to V_j and U_j update to:

$$\frac{\partial L}{\partial U_j} = \frac{1}{J} g_j X^\top (\delta V_j \Sigma_j^\top) - \frac{2\lambda}{Jr} (I + \log(\bar{\rho}_i^L)) U_j \quad (23)$$

And,

$$\frac{\partial L}{\partial V_j} = \frac{1}{J} g_j \delta^\top (H_j \Sigma_j) - \frac{2\lambda}{Jr} (I + \log(\bar{\rho}_i^R)) V_j \quad (24)$$

10 Relationship to gradient subspace methods

The parametrization (1) implicitly induces gradient subspace structure analogous to GaLore/GrassWalk, but without an auxiliary projector. From (5), the gradient seen by U_j is

$$\frac{\partial L}{\partial U_j} \propto X^\top \delta V_j \Sigma_j^\top,$$

i.e., the full upstream signal $X^\top \delta = \partial L / \partial W$ projected onto the right-subspace V_j and rescaled by Σ_j . The role played by an explicit projector S_t in gradient subspace methods is played here by V_j itself: each component absorbs the slice of the gradient along its own right subspace, and the diversity penalty on ρ^R ensures these slices span complementary directions. In this sense the framework unifies weight-side and gradient-side compression: gradient subspace structure is a consequence of the weight parametrization rather than an additional design choice.

Note : We should expand this to explicitly show that GrassWalk/Jump, WASI, and other methods in the space are natural limits or limited cases of the superposition concept we develop here. It shouldn't be too hard, especially for gradient subspace work, but it should be done to really emphasize the theory as flexible, novel, and to highlight the von Neumann entropy on Gr and nonlinear $\phi(u)$ as meaningful theoretical contributions.

11 Extension to multi-head attention

A multi-head attention block has four weight matrices: $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_h H}$ and $W_O \in \mathbb{R}^{d_h H \times d}$, where d is the model dimension, d_h the per-head dimension, and H the number of heads. The forward pass is

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad \text{Attn}(X) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) V W_O.$$

Direct application. Each of W_Q, W_K, W_V, W_O is a linear layer and admits the superposition (1) unchanged. The forward and backward of §5, the Riemannian step of §6, and the diversity penalty of §9 apply per projection without modification. Per-projection forward cost is $O(JBdr)$, favorable when $Jr \ll d$. The softmax and the $QK^\top$ contraction do not see the parametrization: upstream signal arrives at each projection output, and the per-component gradients (5)–(7) decouple as before.

Heads as components. The natural choice for attention is

$$J = H, \quad r = d_h, \quad (25)$$

so that each component of the superposition corresponds to one attention head. The parametrization

$$W_Q = \frac{1}{H} \sum_{h=1}^H f_h^Q(X) U_h^Q \Sigma_h^Q (V_h^Q)^\top, \quad U_h^Q \in \text{St}(d_h, d), \quad V_h^Q \in \text{St}(d_h, d),$$

and likewise for W_K, W_V , is then a constrained reparametrization of multi-head attention itself. Under this identification:

- (i) The diversity penalty $-S(\bar{\rho}^L)$ becomes an explicit *head-diversity* regularizer, pushing different heads to attend to complementary left subspaces of the input. This targets the well-documented head-collapse failure mode in transformers.
- (ii) The content gate $f_h^Q(X) = \phi(\|(V_h^Q)^\top X^\top\|_F^2 / \|X\|_F^2)$ becomes a *per-head input-dependent activation*: head h fires when the input has energy in head h 's right subspace. This is a form of conditional computation tied to head geometry.
- (iii) The effective number of active heads $\text{ENC}_i = \exp(S(\bar{\rho}_i)) / r$ provides a continuous head-utilization diagnostic and a candidate signal for adaptive head pruning.

The structural constraint is that each head's projection is now rank- d_h with an explicit Stiefel-Stiefel factorization, rather than an arbitrary $d \times d_h$ matrix. This is a real restriction relative to unconstrained multi-head attention, but is in line with empirical observations that attention projections are effectively low-rank in trained transformers.

Bilinear form $W_Q W_K^\top$. Attention uses Q and K only through the score matrix $QK^\top = X(W_Q W_K^\top)X^\top$. The effective parameter is the bilinear form $M := W_Q W_K^\top \in \mathbb{R}^{d \times d}$. Under the superposition parametrization,

$$M = \frac{1}{J^2} \sum_{j,k} f_j^Q f_k^K U_j^Q \Sigma_j^Q (V_j^Q)^\top V_k^K \Sigma_k^K (U_k^K)^\top, \quad (26)$$

where $(V_j^Q)^\top V_k^K \in \mathbb{R}^{r \times r}$ is a small alignment matrix between query and key right-subspaces. An alternative parametrization is to model M directly as a superposition rather than W_Q and W_K separately; this couples the query-key diversity into a single regularizer at the cost of losing the per-projection interpretation. We leave this as a design alternative.

Further simplification disbanding the distinction between Q and K would be:

$$M' = \frac{1}{J} \sum_j f_j U_j \Sigma_j V_j^\top, \quad (27)$$

This is likely too extreme a compression, and it loses the tidy separability and interpretability tooling of the linear and bilinear forms discussed above, but is worth at least mentioning.

Caveats.

- (a) *Softmax breaks per-component decoupling for score-level superpositions.* The decoupling argument in §5 relied on the forward being a sum over j with a shared upstream signal. This holds for superpositions of W_Q, W_K, W_V, W_O individually. It does *not* hold for a superposition applied to the score matrix $QK^\top$ before softmax: the softmax Jacobian

$$\frac{\partial \text{softmax}(s)_i}{\partial s_k} = \text{softmax}(s)_i (\delta_{ik} - \text{softmax}(s)_k)$$

introduces cross-component coupling. Chain rule still applies, but per-component parallelism is lost.

- (b) *Rotary position embeddings.* RoPE applies position-dependent rotations to Q and K after projection but before the dot product, mixing within the rank- r output. The interpretation of U_j^Q as picking out a stable feature subspace of X is preserved only up to RoPE rotation; the data-fit loss handles this correctly, but the geometric reading is partial.
- (c) *KV cache.* During autoregressive inference, transformers cache K and V across decoded tokens. Under the superposition, the natural cached objects are the per-component activations $H_j^K = XU_j^K \in \mathbb{R}^{B \times r}$ rather than the full $K \in \mathbb{R}^{B \times d_h H}$, reducing cache size by a factor of $d/(Jr)$. Standard attention kernels (FlashAttention, paged attention) assume the contiguous K, V layout and require adaptation; this is engineering rather than mathematics.
- (d) *Asymptotic complexity.* The superposition reduces the projection cost from $O(Bd^2)$ to $O(JBdr)$, but attention’s dominant cost in long-context settings is the $O(N^2 d_h)$ score computation, which is unaffected. The framework offers parameter and projection-memory savings, not asymptotic attention speedup.

12 Initialization

We initialize $U_{i,j}$ Haar-uniform on $\text{St}(r, n)$ (draw $G \sim \mathcal{N}(0, I)^{n \times r}$, take $U = Q$ from thin QR $G = QR$), and similarly $V_{i,j}$ on $\text{St}(r, m)$. Spectra are initialized to $\Sigma_{i,j} = \sigma_0 I_r$.

Proposition 3 (Variance matching). *With $U_{i,j}, V_{i,j}$ independent Haar-uniform on their respective Stiefels, $\Sigma_{i,j} = \sigma_0 I_r$, and $f_{i,j} \equiv 1$ at initialization, the entries of $\widetilde{W}_i$ have mean zero and*

$$\text{Var}(\widetilde{W}_{i,ab}) = \frac{r \sigma_0^2}{J n m} (n + m + O(1)).$$

For square layers $n = m$, this simplifies to $\text{Var}(\widetilde{W}_{i,ab}) = 2r\sigma_0^2/(Jn) + O(n^{-2})$. To match Kaiming initialization ($\text{Var} = 2/n$, for ReLU networks), set

$$\sigma_0^2 = \frac{J}{r}.$$

Proof sketch. For Haar-uniform $U \in \text{St}(r, n)$, $\mathbb{E}[UU^\top] = (r/n)I_n$. Independence of U_j, V_j, U_k, V_k across j and sides gives mean zero and a variance computation by expanding $\widetilde{W}_{ab} = (1/J) \sum_j \sigma_0 \sum_k U_{j,ak} V_{j,bk}$. $\square$

Remark 4. *Products of orthonormal-times-Gaussian-like matrices have heavier tails than a fresh Gaussian; the second moment matches Kaiming/Xavier but the tails do not. Empirically this is unlikely to matter at init but is worth noting.*

Remark 5. In high dimension ($r \ll n$), Haar-uniform draws on $\text{St}(r, n)$ already yield near-orthogonal subspaces across j in expectation: $\mathbb{E}\|U_j^\top U_k\|_F^2 = r^2/n$ for $j \neq k$, which is small. Explicit rotation to enforce orthogonality at init is therefore unnecessary; the entropy regularizer maintains it during training.

13 Training

Remark 6. While a simple first pass implementation will probably feature lots of sequential calculations, in theory there is **massive** potential for parallelization or sequence breaking across the method. If we intelligently harness this in a final version, I think a $\times 10$ training acceleration is realistic. The question will be if the results are competitive, and if we can do interesting interpretability work.

14 Open questions and next steps

1. **Identifiability under the content gate.** The gate (3) breaks the rank- Jr equivalence by tying activation to V_j . Establishing that the gated decomposition is identifiable (up to the residual $O(r)$ gauge per component) under reasonable data distributions is open.
2. **Optimal ϕ .** The choice of ϕ in (3) is currently heuristic. A natural candidate is to derive ϕ from a maximum-entropy principle: among gate functions producing fixed marginal activation rates, which maximizes the entropy of the component assignment? Another choice is my position (below), softmax, because it has attractive information geometry and connections to MoE/Information Theory work.
3. **Adaptive J .** The effective-number-of-components ENC_i in §9 could drive a pruning/growing scheme on J per layer and over time with periodic regrounding.
4. **Empirical comparison to $J = 1$, rank Jr .** The cleanest test of whether the gated $J > 1$ parametrization buys anything over a single rank- Jr factorization. it will be important to show that we can reach WASI and GrassWalk or other activation/gradient subspace methods as consequences of this system - but a core difference is that we have an entropy factor on $\text{Gr}()$ that they do not use.
5. **Phase variants.** Complex-coefficient extensions $c_j = \sqrt{p_j}e^{i\phi_j}$ are a natural avenue for additional gradient-flow modulation but require either a complex-valued forward path or a structured real lift; deferred.

15 Junk & Notes

1. **Optimal J ?** How can we find an optimal J ? We could dynamically estimate it, having some $J = L(i, X, \theta)$ function clamped in a feasible range for compute? Maybe it would have something to do with the span of the subspaces relative to $\mathbb{R}^{n \times m}$? I had been imagining that our r values might be at most $\mathcal{O}(1000)$, and J being 4 to 16 at most, but that's not based on any calculation on results.
2. **Interpretability of $U_j \Sigma_j V_j^T$:** If $\phi(X)$ follows some of the proposed pathways, then the dominant subspaces for a given X should be easy to identify, and repeatably identifiable at

test time later. We could do this to massively reduce the overhead of the influence function H^{-1} calculation from $\mathcal{O}(n^3)$ to $\mathcal{O}(r^3)$, and we could then approximate the impact in other subspaces using $\phi(u)$.

3. **Misc:** Can we steer models, detect high and low value dataset samples, or do behavior manifold stuff in these subspaces at train and/or test time? Could be very interesting.
4. **Experiments:** Obviously we want to measure training speed, memory, and quality vs dense for LLM pretraining here. Other methods might be good to compare to, but will be tricky and slow to re-implement in-house.
5. **Multihead Attention:** The big kahuna, how the hell do we extend this to MHA? I highly doubt we can claim any superior throughput complexity in aggregate or at test time, but maybe we can frame the contribution as a solution to head diversity collapse? A naive idea here is to just swap Q, K, V in for W in the above formalism, but I'm not sure the gauge survives that?
6. **[Revised] Position on $\phi(u)$:**

I suggest we use SoftMax over J , $\phi(u) = \text{Softmax}(u)$, so we can gain access to attractive information geometry tools from MoE literature and information theory. I think this gives us the strongest path to interpretability applications. The consequences of this choice are also closer in spirit to the quantum mechanical inspiration for potentially choosing complex/phase space $\phi(u)$, which is neat. A tradeoff here is that a single sample u has nonzero contribution across all J subspaces, whereas with something like $\arg \max\{\text{Softmax}(u_1), \dots, \text{Softmax}(u_j), \dots, \text{Softmax}(u_J)\}$ confines the contribution of u to a single subspace. This is probably too naïve, but is probably also worth mentioning?

However, it's probably fine to use $\phi(u) = u$ or $\phi(u) = e^u$, for simplicity as well as gradient flow. As these $\phi(u)$ candidates have no learned components.

Alternative future options might be complex phases, something like $\phi(u) = \exp(-\alpha u i)$ and allowing the subspaces to interfere in the complex plane. This might have the effect of working against the density matrix derived diversity penalty in $\text{Gr}(r, n)$, and would add overhead from complex backward passes (obviously the forward passes need to happen in $\text{Re}(\theta X)$). I am not sure how well torch or JAX support complex tensor calculus, and am not eager to find out the hard way.

7. **QR Retraction vs Geodesics** QR retraction is a decent approximation, but the Geodesics on $\text{Gr}(r, n)$ and $\text{St}(r, n)$ are known - GrassWalk/Jump use them. We could swap to them, but there's a tradeoff with complexity/time and it's impacts on the periodic regrounding that needs to be understood to decide intelligently there.

16 Future Directions and Extensions

1. **Minkowski Spaces in Diffusion Models:** Diffusion models have the minkowski representation hypothesis, a time dependent coupling of the manifold representation hypothesis. If we extend our geometry to Minkowski spaces, we might be able to work on these models (text, image, video/audio, etc) in a similar way. There might even be a chance to unify the formalisms as intrinsic vs extrinsic time coordinations of the same geometry.

2. **Theory of Interpretability:** If we have promising results for interpretability of subspaces, perhaps we can use the geometry to formalize a more general theory of interpretability? This is super hazy but would be fucking dank if we pulled it off. Perhaps an angle would be to go from behavior manifolds to behavior topologies, or do some algebraic geometry on the set of behavior manifolds captured by the subspaces? Still, this is very fuzzy.

I am doing a *lot* of reading and studying to catch up with what methods are promising/popular here, but honestly I'd be open to bringing in another collaborator to help us with some extra domain knowledge here.

3. **Proactive Interpretability:** This is a longshot, but by tweaking the gating function $\phi(u)$ we could *a priori* determine what influence gets put in which subspace(s). I highly doubt this would work well, so maybe it's best left to future work, but it would be a real contribution to show that you can bake interpretability into a model from initialization, rather than trying to recover it after training.

Remark 7. Suppose we do admit complex gates of the form $\phi(u) = \alpha \exp(-\beta u i)$. The consequence of this choice is that the forward process happens with $\tilde{A}_i \in \mathbb{C}^{n \times m}$ for layers $i \neq L \in L$. While the the output of this system would need to be $\text{Re}(\tilde{A}_L)$, backpropagation could happen in the complex phase space. This may be attractive in that it would allow the subspaces j to destructively interfere natively through α and β .

However, I am unsure of the impact this would have on the broader geometry, unless $\text{Gr}()$ and $\text{St}()$ are well behaved. Another question would be the support of PyTorch or JAX (I refuse to use TensorFlow on principle) for this system. Furthermore, it's unclear if the native complex interference between subspaces is superior to the density matrix method outlined here. This may remain a curious direction for future work.

17 Simplified Early Build

A sketch for a simplification as an early prototype. The idea is that if this does not work, then the more advanced method above will definitely not work. I cannot emphasize enough that this method sacrifices a *lot* to be as quickly prototype-able as possible. We can start with an arbitrary $J = 4$ and go from there.

$$W \approx \widetilde{W} = \frac{1}{J} \sum_{j \in J} f_j U_j \Sigma V_j^T + b_j$$

We can recycle the Haar Uniform initializations, gating functions $\phi(u)$, and skip the regroundings, manifold steps, optimizer adaptation (or just use SGD to test), von Neumann entropy, and other tricks for simplicity. To avoid degenerate subspaces, we can just say that,

$$\nabla J_{j \in J} \perp \nabla J_{k \in J, k \neq j}$$

We can do this projection quickly with the Haar Uniform initialization method for a quick orthogonal projection. This way we can dodge doing these gradient adjustments sequentially for $j \in J$.

This method accepts many limitations, but is fairly quickly build-able for quick and dirty conceptual verification. I like the MoE style gate, so we can use:

$$f_j(X) = \text{Softmax} \left(\frac{\|V^T X^\top\|_F^2}{\|X\|_F^2} \right)$$

For a simplest possible starting point, we can use a simple MLP on MNIST or Fashion MNIST as a testbed. It's quick, laptop friendly, and easy to tinker with. This is also a great place to test some interpretability ideas and make pretty visuals. For example, with $J = 10$ you would expect each subspace might focus on a specific number, or with $J = 7$ each subspace might learn features similar to a seven-segment display.

Remark 8. *Update:* *this does not work because $U\Sigma V^T$ do not maintain their correct forms without persistent maintenance. However when we introduce this maintenance, we start to arrive at the above form anyways. My assessment that if this method did not work then the above would definitely not was probably wrong. I am brainstorming other ways to test this.*